

R. Bernadett Vlocskó

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Professional Summary

- PhD-trained medicinal chemist with 8+ years of experience in synthesis, method development, and molecular modeling
- Award-winning researcher across 3 countries; recognized as top-performing chemistry graduate student at UMass Boston
- Designed small molecule series and advanced compounds through biological evaluation and lead selection, resulting in novel IP and multiple peer-reviewed publications in leading journals
- Pharmaceutical industry experience in CMC at Agios and formulation development at Hovione
- No visa sponsorship required; OPT approved with path to permanent residency

Technical Skills

Medicinal & Synthetic Chemistry:	Small molecule design, analog library synthesis, SAR analysis, synthetic route design, multi-step synthesis, reaction development and optimization, purification
Analytical & Characterization:	NMR, LC-MS, GC-MS, HPLC, flash chromatography, XRPD, UV-Vis, FTIR, TGA, DSC
Computational Chemistry:	Molecular docking (Schrödinger), molecular/quantum mechanics (Gaussian, ORCA), physicochemical property analysis (COSMOtherm), crystal structure analysis (Mercury)
Programming & AI:	Python, Bash scripting, AI-assisted scientific workflows

Experience

University of Massachusetts Boston <i>Postdoctoral Research Associate</i>	Boston, MA May 2026–Present
• Developing a ^{19}F NMR-based reaction profiling workflow to characterize heat release during chemical synthesis	
University of Massachusetts Boston <i>PhD Researcher, Organic Chemistry</i>	Boston, MA Jan 2020–May 2026
• Designed and synthesized a hydrazone antioxidant library for preeclampsia therapy, using SAR and computational analysis to identify structural drivers of activity and prioritize candidates	
• Advanced selected compounds into cell-based testing, identified a lead candidate, and used the results to design second-generation analogs with improved mitochondrial targeting and drug-like properties	
• Developed microwave-assisted synthetic routes for hydrazone program, reducing reaction times and achieving >95% purity	
• Developed a metal-free and acid-free route to benzimidazole / benzothiazole antioxidants, achieving near-quantitative yields	
• Mentored ~10 lab trainees and taught organic chemistry coursework for five years (~10 modules, >200 students)	
• Research outputs: novel IP (WO2026174168 A1), 7 manuscripts, and presentations at major chemistry conferences	
Agios Pharmaceuticals <i>Graduate Research Intern, Drug Substance Group, CMC</i>	Cambridge, MA Jun–Aug 2024
• Built a computational solid form screening workflow for salts and cocrystals, combining molecular modeling, crystal structure analysis, and experimental characterization	
• Tested the workflow in a proof-of-concept study using on-market antihistamine APIs, identifying two novel salt forms	
• Presented the project to executive leadership as an example of CMC's computational capabilities	
Hovione <i>Research Scientist Intern, Inhalation & Advanced Drug Delivery</i>	Lisbon, Portugal Aug–Dec 2019
• Developed and optimized a dry-powder liposomal formulation of fluticasone propionate for sustained lung delivery	
• Optimized high-pressure homogenization and spray-drying processes to produce reproducible, controlled particle properties	
Budapest University of Technology and Economics <i>MSc Researcher, Organic Chemistry</i>	Budapest, Hungary 2017–2019
• Designed and synthesized azobenzene molecular switches and optimized their immobilization on composite surfaces	

Education

University of Massachusetts Boston <i>PhD, Organic Chemistry</i> <i>Béla Török Research Group</i> Thesis: Computer-aided Development of Novel Antioxidants	Boston, MA Jan 2020–May 2026
Budapest University of Technology and Economics <i>MSc, Chemical Engineering, Drug Development and Manufacturing</i> <i>Functional Organic Materials Group</i> Thesis: Dynamically Controllable Interfacial Systems Containing Photoactive Azobenzenes	Budapest, Hungary 2017–2019
University of Szeged <i>BSc, Chemistry</i> Thesis: Cinchona Alkaloid-Catalyzed Asymmetric Michael Addition of Acetylbutyrolactone to Activated Olefins	Szeged, Hungary 2014–2017

Publications

1. Zsengellér, K. Zs., Mastuyugin, M., Fusco, R. A., **Vlocskó, R. B.**, et al. Impact of Structural Features on the Antioxidant Activity of Organofluorine Diaryl Hydrazones. *Molecules* **2026**
2. **Vlocskó, R. B.**, et al. Development of Diaryl Hydrazones for Alleviation of Mitochondrial Oxidative Stress in Preeclampsia. *J. Med. Chem.* **2025**
3. Zsengellér, K. Zs., Mastuyugin, M., Fusco, R. A., **Vlocskó, R. B.**, et al. Mitigating Oxidative Stress and Anti-Angiogenic State in an In Vitro Model of Preeclampsia by HY-12, an Organofluorine Hydrazone Antioxidant. *Curr. Issues Mol. Biol.* **2025**
4. **Vlocskó, R. B.**, et al. Correlation of Physicochemical Properties with Antioxidant Activity in Phenol and Thiophenol Analogues. *Sci. Rep.* **2025**
5. **Vlocskó, R. B.**, et al. Catalyst-free Synthesis of Substituted Benzimidazoles and Benzothiazoles in a Sustainable Solvent. *Tetrahedron Green Chem.* **2024**
6. **Vlocskó, R. B.**, et al. Green Synthesis of Aromatic Nitrogen-Containing Heterocycles by Catalytic and Non-Traditional Activation Methods. *Molecules* **2023**
7. Schäfer, C., Cho, H., **Vlocskó, R. B.**, et al. Recent Advances in the Green Synthesis of Heterocycles: From Building Blocks to Biologically Active Compounds. *Curr. Org. Synth.* **2022**
8. Kunfi, A., **Vlocskó, R. B.**, et al. Photoswitchable Macroscopic Solid Surfaces Based on Azobenzene-Functionalized Polydopamine/Gold Nanoparticle Composite Materials. *ChemPlusChem* **2020**

Select Conference Presentations

1. **Vlocskó, R. B.**, et al. Exploring Antioxidant Effects of Organofluorine Diphenyl Hydrazone Quaternary Salts in Preeclampsia Therapy: a DFT Approach. *ACS Spring Meeting* **2025**
2. **Vlocskó, R. B.**, et al. Catalyst-free Approaches in Green Synthesis: Environmentally Benign Synthesis of Benzimidazoles and Benzothiazoles in a Sustainable Solvent. *GC&E Conference* **2024**
3. **Vlocskó, R. B.**, et al. Assessing Antioxidant Activity Through Theory and Experiments: A Comparative DFT Study of Phenol and Thiophenol Analogues. *ACS Spring Meeting* **2024**

Awards

- **Vincent T. Mastricola, Jr. Graduate Fellowship:** top-performing graduate student, UMB Chemistry Department (2025)
- **CSM Dean's Doctoral Research Fellowship:** UMB College of Science & Mathematics (2025)
- **Ciba Green Chemistry Award:** ACS Green Chemistry Institute (2024)
- **Graduate Teaching Award:** UMB Chemistry Department (2021)
- **Ivan Villax Scholarship:** Hovione, Portugal (2019)
- **New National Excellence Program Scholarship:** National Research and Innovation Office of Hungary (2016–2017)